

Fraud Detection System - Major Classes and Their Purpose

1. MultiFeatureCategoricalFraudTransformer

Purpose: Transforms categorical features in a fraud detection dataset by encoding categories that show a strong relationship with fraud.

Key Parameters:

- `category_cols`: List of categorical columns to transform.
- `label_col`: The name of the target column (usually indicating fraud).
- `top_k`: Number of top categories to keep based on fraud signal.
- `fraud_coverage`: Threshold of fraud coverage for inclusion of a category.

Key Methods:

- `fit`: Analyzes the relationship between each category in `category_cols` and the fraud label.
- `transform`: Transforms the dataset by encoding categories that meet the fraud coverage or `top_k` threshold.
- `fit_transform`: Combines `fit` and `transform`.

Use Case: Used after loading the data and selecting categorical columns. Ensures features are transformed before rule mining or training.

2. RuleExtractor (Assumed)

Purpose: Extracts interpretable rules from the dataset based on tree-based models such as XGBoost.

Key Parameters:

- `feature_cols`: Columns used to extract decision rules.
- `target_col`: Label used to train the underlying model.
- `tree_params`: Dictionary of XGBoost parameters.

Key Methods:

- `extract_rules`: Trains a model and extracts paths (rules) from decision trees.
- `apply_rules`: Filters the dataset using extracted rules.

Use Case: Invoked after feature selection to simulate fraud investigation via decision rule mining.

3. RuleBuilder (Assumed)

Purpose: A utility class to manage and combine multiple rules logically.

Key Methods:

- `add_condition`: Adds a new condition to the rule set.
- `clear_conditions`: Resets all current conditions.
- `evaluate`: Applies the current rule set to a dataset.

Use Case: Used for managing user interactions around rule creation or iteration-based refinement.

4. SessionState (Streamlit-native concept)

Purpose: Acts as a memory layer across reruns. Maintains state for:

- DataFrames (`original_df`, `current_df`)
- Applied rules
- Current iteration
- Selected features and model parameters

Use Case: Essential for preserving the app's workflow and avoiding restarts due to user inputs.

Summary:

These classes and constructs work together to load fraud datasets, encode meaningful patterns in categorical variables, and allow analysts to mine interpretable rules. The design aims to support iterative, business-friendly fraud investigation via human-in-the-loop workflows.